MIL-P-23929A
12 October 1964

Superseding
MIL-P-23929(Wep)
12 November 1963

MILITARY SPECIFICATION

PROPELLANT, H8

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the basic requirements for a special type of propellant that is produced as a cylindrical single perforated grain.

1.2 Classification - Unless otherwise specified, the propellant shall be manufactured in three grain sizes (see 6.2):

(1) Grain A
(2) Grain B
(3) Grain C

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-G-491	Glycerol, Technical (High Gravity)

Military

MIL-D-204	Dinitrotoluene (For Use in Explosives)
MIL-C-206	Cellulose, Cotton (For Use in Explosives)
MIL-N-244	Nitrocellulose
MIL-N-246	Nitroglycerin
MIL-E-255	Ethyl Centralite (Carbamite)
MIL-P-270	Propellant, Artillery
MIL-T-301	Triacetin (Glyceryl Triacetate)
MIL-L-758	Lead Stearate (For Use in Ammunition)

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:22Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-286	Propellants, Solid; Sampling, Examination, and Testing
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

Bureau of Naval Weapons

138439	Packing Box, Mark VII Mod 0, for Smokeless Powders
593131	Packing Box, Mark VII Mod 1, for Smokeless Powders
1350557	Packing Box, Mark VII Mod 2, for Smokeless Powders
1380388	Packing Box, Mark VII Mod 3, for Smokeless Powders
561968	Powder Container, Mark 1 Mod 0

PUBLICATIONS

Bureau of Naval Weapons

NAVORD OD 9375	Standard Methods and Procedures for the Determination of Heat of Explosion of Rocket Propellant Powders

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publications should cite the latest edition and supplements thereto.

3. REQUIREMENTS

* 3.1 Data - No data is required by this specification (other than the lot acceptance data specified in 3.10) or by the applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:22Z
Check the source to verify that this is the current version before use.

3.2 Materials - The raw materials used in the manufacture of the propellant shall conform to the applicable specifications listed in Table I.

3.2.1 Composition - After testing as specified in 4.5, the composition of the propellant, exclusive of moisture and lead stearate, shall be in accordance with Table I.

TABLE I

COMPONENT	SPECIFICATION	PERCENT
Nitrocellulose	MIL-N-244 1/ Type 2, Grade C	58.50 ± 1.00
Nitroglycerin	MIL-N-246 1/ Type I	22.50 ± 0.50
Dinitrotoluene	MIL-D-204	2.50 ± .25
Ethyl Centralite	MIL-E-255, Class II	8.00 ± .30
Triacetin	MIL-T-301	8.50 ± .35

* 1/ Unless otherwise specified in the contract or order, the cellulose used in the manufacture of the nitrocellulose shall be derived from cotton linters and conform to MIL-C-206; and the glycerin used in the manufacture of nitroglycerin shall be synthetically produced and conform to O-G-491. (see 6.2)

3.2.1.1 Lead stearate - Lead stearate conforming to MIL-L-758 shall be added to the propellant. The lead stearate content shall be not less than 0.45 percent nor more than 0.50 percent when tested as specified in 4.5.

* 3.2.1.2 Moisture - The moisture content of the propellant shall be not more than 0.40 percent when tested as specified in 4.5.

3.2.1.3 Ash - The ash content of the propellant shall be not more than 0.40 percent when tested as specified in 4.5.

3.3 Specific gravity - The specific gravity of the propellant shall be not less than 1.515 nor more than 1.555, when tested as specified in 4.5.

3.4 120°C heat test - When tested as specified in 4.5 the propellant shall not cause complete fading of the methyl violet paper to salmon pink in less than 40 minutes and shall not explode in less than 5 hours.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:22Z
Check the source to verify that this is the current version before use.

3.5 Form and dimensions of grains -

3.5.1 Form - The grain form shall be cylindrical with a single perforation at the center of the grain.

* 3.5.2 Nominal grain dimensions - Unless otherwise specified in the contract or order, there shall be three grain sizes. Nominal grain dimensions, when measured as specified in 4.5.1, shall conform to the requirements of Table II. The nominal dimensions of the grain can be changed when loading conditions or cartridge type so requires. The particular dimensions required shall be given in the contract or as an addendum to this specification. (see 6.2)

TABLE II

	DIMENSIONS IN INCHES		
DIMENSION	Grain A	Grain B	Grain C
Length	0.440	1.040	2.700
Outside diameter	0.178	0.370	0.318
Perforation diameter	0.018	0.130	0.088
Web thickness	0.080	0.120	0.115

* 3.5.3 Length and diameter uniformity - The length and diameter of the grain shall conform to the uniformity requirements of Table III.

TABLE III

STANDARD DEVIATION 1/ OF INDIVIDUAL DIMENSIONS FROM THE MEAN DIMENSION, EXPRESSED AS PERCENTAGE OF THE MEAN DIMENSION

DIMENSION	MAXIMUM
Length	6.25 percent
Diameter (greater than 0.2 inch)	3.125 percent
Diameter (less than 0.2 inch)	6.25 percent

1/ The standard deviation shall be calculated by a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, example B-1.

* 3.5.4 Web measurement - The average of the web measurements above the median shall not differ from the median by more than 15 percent.

3.6 Heat of explosion - When determined as specified in 4.6, the heat of explosion shall be not less than 753 nor more than 778 calories per gram.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:22Z
Check the source to verify that this is the current version before use.

3.7 Additional detail requirements - Any ballistics, or other detail requirements pertaining to the propellants intended use or application, to grain characteristics, fissures, foreign material or other physical or chemical properties shall be specified in the applicable addendum or contract (see 6.2).

3.8 Process approval - Details of the manufacturing process and equipment used by the contractor shall be submitted in writing to the contracting agency and shall be approved before the contractor proceeds with the manufacture and packing of the propellant. Any deviation from the approved manufacturing process shall be submitted in writing to the contracting agency and written approval received prior to use (see 6.2).

* 3.9 Reworked material - The finished propellant (each grain size) may contain a uniform percentage of rework material. Rework material shall consist of clean scrap such as clippings, ends, mill chips, and whole and sectional grains. Magnetic separators shall be used to remove any possible metal contamination from the rework material. Prior to use, the contractor shall furnish evidence of the satisfactory stability of the rework material (see 6.2).

3.10 Description sheet - With every lot of propellant submitted for acceptance inspection, the contractor shall furnish, on official forms, eight copies of a description sheet giving a complete history of its manufacture and the results of all chemical and physical analyses and examinations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility - Unless otherwise specified by the Government, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the examinations and tests set forth in the specification where such are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection - Sampling and inspection shall be in accordance with MIL-P-270, unless otherwise specified herein.

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:22Z
Check the source to verify that this is the current version before use.

4.3 Lot formation - For purposes of inspection, a lot shall consist of 40,000 pounds maximum and shall be formed in accordance with MIL-STD-105. Each lot, as far is practicable, shall consist of units of product that have been manufactured under essentially the same conditions and at essentially the same time.

4.4 Sampling -

4.4.1 Chemical, Stability and Surveillance Samples - Ten pounds of propellant shall be taken at random from each lot from the larger ends cut from a minimum of 50 sticks of propellant during the milling operation. Five pounds shall be used for chemical and stability tests and the remaining five pounds shall be forwarded to a specified government laboratory for 65.5°C surveillance tests (see 6.2). All samples shall be packed in clean airtight containers and shall be marked to show the propellant designation, lot number, manufacturer, date of sampling, contract number, and number of pounds in the lot.

4.4.2 Ballistics Samples - If the contract requires that a ballistics or performance test be made, a sample of appropriate size and kind as specified in the contract shall be sent to the procuring activity or to the activity designated by the procuring activity (see 6.2).

* 4.4.3 Samples for examination of filled containers - Sampling plans and procedures for the examination of filled containers shall be in accordance with MIL-STD-105 except that 100% of the product shall be inspected for a critical defect when listed by the contracting agency, with zero defects acceptable. (see 6.2)

4.5 Test methods - Tests shall be conducted in accordance with MIL-STD-286 using the applicable test methods listed in Table IV and 4.5.1.

*

TABLE IV

	METHOD NO.
Nitrocellulose	209.2
Nitroglycerin	208.1 1/
Dinitrotoluene	205.2
Ethyl Centralite	202.2
Triacetin	*207.5
Lead stearate	*311.1
Moisture	102.1
Ash	106.4
Specific gravity	510.1
120°C heat test	404.1

1/ Use alcohol-free anhydrous either or methylene chloride for the sample extraction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:22Z
Check the source to verify that this is the current version before use.

4.5.1 Grain dimensions - Thirty grains shall be selected from each lot for examination. Measurements shall be made in accordance with method 504.1 of MIL-STD-286.

4.6 Heat of explosion - The heat of explosion shall be determined by the method given in NAVORD OD9375, Standard Methods and Procedures for the Determination of Heat of Explosion of Rocket Propellant Powders.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Not applicable.

5.2 Packing -

5.2.1 Level A - In addition to the requirements of 5.2.1.1, packing shall be in accordance with the Code of Federal Regulations 49CFR71-78.

5.2.1.1 Exterior containers - Propellant shall be packed in metal boxes conforming to Navy Bureau of Ordnance Drawings 138439, 593131, 1350557 and 1380388, (MK VII Mod 0, MK VII Mod 1, MK VII Mod 2, and MK VII Mod 3, respectively) or to Navy Bureau of Ordnance Drawing 561968 (MK 1 Mod 0) as specified in the contract or purchase order (see 6.2). Immediately prior to packing, all containers shall be tested with compressed air; new containers at an internal pressure of 3 pounds per square inch, old containers at an internal pressure of 5 psi. The containers shall withstand the specified pressure for 15 seconds without leakage.

5.2.1.2 Levels B and C - Not applicable.

5.3 Marking - In addition to any special marking required by the contract or purchase order (see 6.2), containers shall be marked in accordance with the applicable drawings specified in 5.2.1.1, MIL-STD-129 and the Code of Federal Regulations 49CFR71-78. The lot number shall also be marked on each container.

6. NOTES

6.1 Intended use - The propellant covered by this specification is intended for use in cartridges for actuated devices.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:22Z
Check the source to verify that this is the current version before use.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Grain size (see 3.5.2).
(c) Ballistic or other detail requirements (see 3.7).
(d) Process approval (see 3.8).
(e) Other requirements for cellulose and glycerin (see 3.2.1, note under Table I)
(f) Inspection and acceptance conditions (see 4.4.3).
(g) Type of container and quantity (see 5.2.1.1).
(h) Special marking requirements (see 5.3).
(i) Information concerning government material furnished.
(j) The method for determining the suitability or stability of the rework material i.e., 120°C heat test or the 65.5°C surviellance test (see 3.9).
(k) Any additional data requirements (see 3.1).
(l) Forwarding address for surveillance or ballistic samples (see 4.4.1 and 4.4.2).

6.3 In case the propellant fails the ballistic or performance tests in the device for which it is intended, the Bureau or agency concerned may; if deemed to the governments advantage, accept the propellant upon its meeting the specifications of the Bureau or agency for use in any other device or weapon.

6.4 All expenses of resubmitted lots shall be borne by the contractor, even if finally accepted for the weapon for which originally intended. Ballistic, chemical, or stability samples, representing finally rejected lots, shall not be paid by the government. Surveillance samples shall be paid for only if the lot is finally accepted.

6.5 Marginal notations - The outside margins of this specification have been marked to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:22Z
Check the source to verify that this is the current version before use.

Custodians:

Army - MU

Navy - WP

Air Force - OOA

Preparing Activity:

Navy - WP

DOD Project Number 1375-0742

Review Activities:

Army - MU
Navy - WP
Air Force - 14

User Activities:

Air Force - 70

Review/user information is current as of the date of this document. For future coordination of changes this document, draft circulation should be based on information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:22Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-P-23929 Propellant, H8 Revision A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:22Z
Check the source to verify that this is the current version before use.